

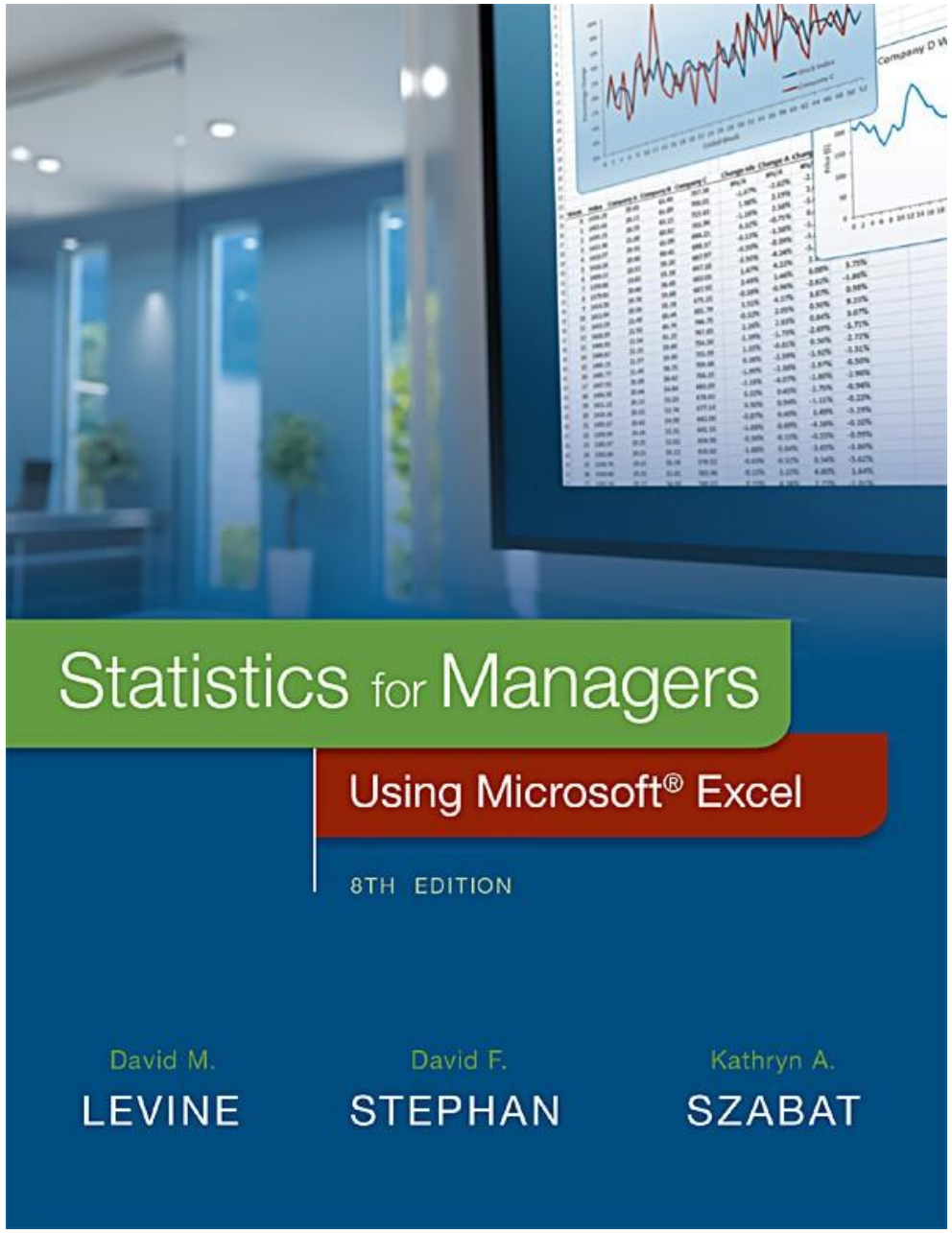
Session 5 Agenda

1. Classroom Management
2. Recap of Session 4
3. Lecture on Discreet Probability Distributions
4. Reminders: Deliverables for Session 6

Quiz Tough Question #9

A box of 9 cellphones contains 2 gold and 7 silver ones. If 2 cellphones are randomly selected from the box, without replacement (the first cellphone is not returned to the box after it is selected), what is the probability that there will be 1 gold cellphone **AND** 1 silver cellphone selected?

a) 0.03 b) 0.39 c) 0.47



Chapter 4
Discreet Probability Distribution

PAUL ANDREW GORGONIO

Learning Objectives

In this chapter, you learn:

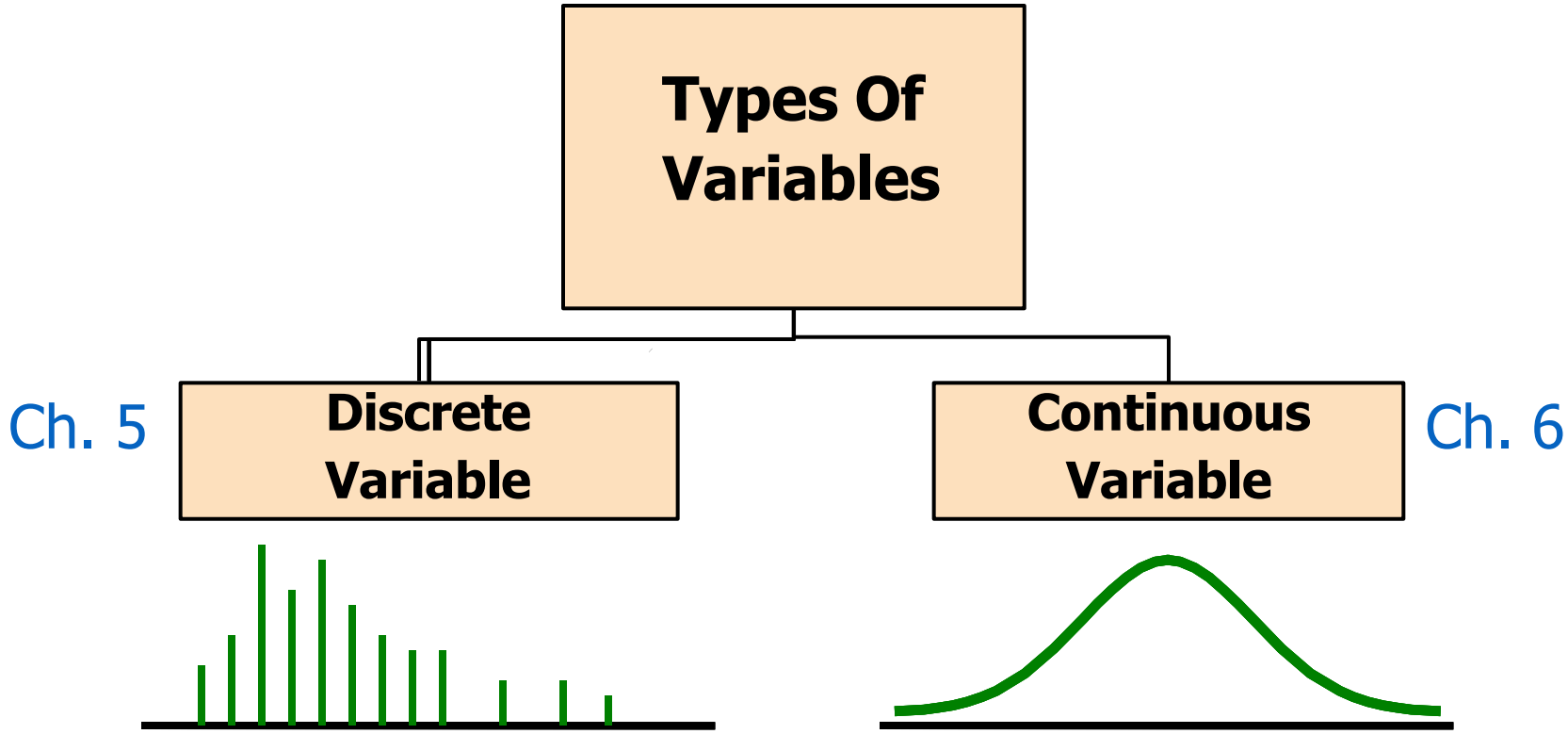
- The properties of a probability distribution
- To compute the expected value
- To compute probabilities from binomial and Poisson distributions

Definitions

- **Discrete variables** produce outcomes that come from a counting process (e.g. number of classes you are taking).

- **Continuous variables** produce outcomes that come from a measurement (e.g. your annual salary, or your weight).

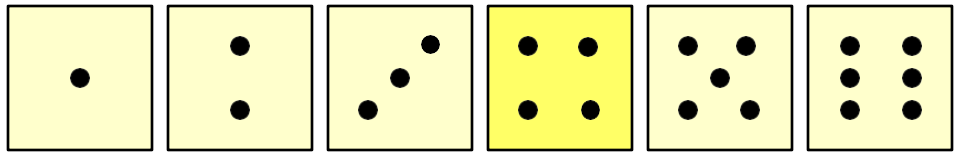
Types Of Variables



Discrete Random Variables

- Can only assume a countable number of values

Examples:



- Roll a die twice

  Let X be the number of times 4 occurs

  (then X could be 0, 1, or 2 times)

- Toss a coin 5 times.

  Let X be the number of heads

  (then X = 0, 1, 2, 3, 4, or 5)



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Probability Distribution For A Discrete Random Variable

- A mutually exclusive listing of all possible numerical outcomes for that variable and a probability of occurrence associated with each outcome.

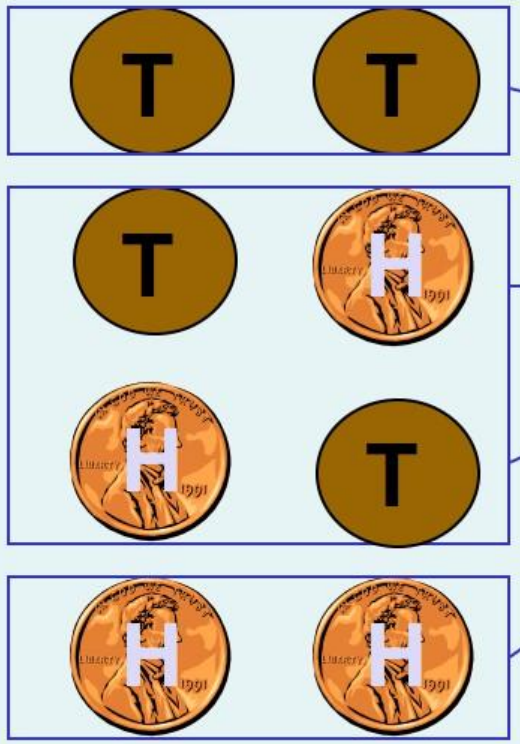
Interruptions Per Day In Computer Network	Probability
0	0.35
1	0.25
2	0.20
3	0.10
4	0.05
5	0.05

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Example of a Discrete Random Variable Probability Distribution

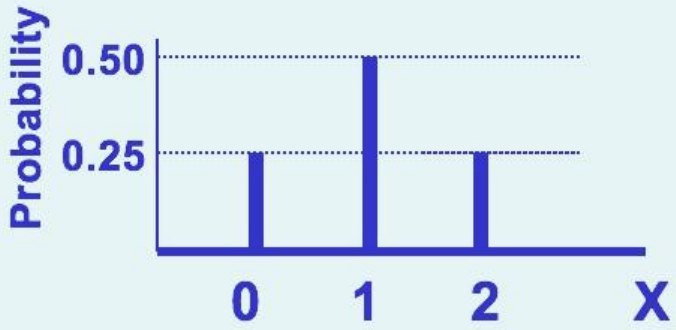
Experiment: Toss 2 Coins. Let X = # heads.

4 possible outcomes



Probability Distribution

X Value	Probability
0	1/4 = 0.25
1	2/4 = 0.50
2	1/4 = 0.25



Discrete Variables
Expected Value (Measuring Center)

Expected Value (or mean) of a Discrete Variable
(Weighted Average)

$$\mu = \mathrm{E}(X) = \sum_{i=1}^{N} X_i P(X = X_i)$$

- Example: Toss 2 coins,
  X = # of heads,
  compute expected value of X:

X	$P(X=X_i)$
0	0.25
1	0.50
2	0.25

E(X) = ((0)(0.25) + (1)(0.50) + (2)(0.25))
= 1.0

Discrete Variables
Expected Value (Measuring Center)

Expected Value (or Mean) of a Discrete Variable

Interruptions Per Day In Computer Network (x_i)	Probability $P(X = x_i)$	$x_i P(X = x_i)$
0	0.35	(0)(0.35) = 0.00
1	0.25	(1)(0.25) = 0.25
2	0.20	(2)(0.20) = 0.40
3	0.10	(3)(0.10) = 0.30
4	0.05	(4)(0.05) = 0.20
5	0.05	(5)(0.05) = 0.25
	1.00	$\mu = E(X) = 1.40$

Discrete Variables: Measuring Dispersion

- Variance of a discrete variable.

$$\sigma^2 = \sum_{i=1}^{N} [x_i - E(X)]^2 P(X = x_i)$$

- Standard Deviation of a discrete variable.

$$\sigma = \sqrt{\sigma^2} = \sqrt{\sum_{i=1}^{N} [x_i - E(X)]^2 P(X = x_i)}$$

where: E(X) = Expected value of the discrete variable X

x_i = the ith outcome of X

$P(X=x_i)$ = Probability of the ith occurrence of X

Discrete Variables: Measuring Dispersion

$$\sigma = \sqrt{\sum_{i=1}^{N} [x_i - E(X)]^2 P(X = x_i)}$$

Interruptions Per Day In Computer Network (x_i)	Probability $P(X = x_i)$	$[x_i - E(X)]^2$	$[x_i - E(X)]^2 P(X = x_i)$
0	0.35	$(0 - 1.4)^2 = 1.96$	$(1.96)(0.35) = 0.686$
1	0.25	$(1 - 1.4)^2 = 0.16$	$(0.16)(0.25) = 0.040$
2	0.20	$(2 - 1.4)^2 = 0.36$	$(0.36)(0.20) = 0.072$
3	0.10	$(3 - 1.4)^2 = 2.56$	$(2.56)(0.10) = 0.256$
4	0.05	$(4 - 1.4)^2 = 6.76$	$(6.76)(0.05) = 0.338$
5	0.05	$(5 - 1.4)^2 = 12.96$	$(12.96)(0.05) = 0.648$
			$\sigma^2 = 2.04$, $\sigma = 1.4283$

Counting Rules 1

COUNTING RULE 1

If any one of k different mutually exclusive and collectively exhaustive events can occur on each of n trials, the number of possible outcomes is equal to

$$k^n \qquad (4.10)$$

- The # of different possible outcomes from tossing a two-sided coin five times:
  $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$
- What is the total # of different outcomes if a six-sided die is rolled twice?
- You take a multiple choice test made up of 10 questions. Each question has 4 possible answers. How many different ways are there to answer the test (assuming you don't leave a question blank)?

Counting Rule 2

COUNTING RULE 2

If there are k_1 events on the first trial, k_2 events on the second trial, . . . , and k_n events on the nth trial, then the number of possible outcomes is

$$(k_1)(k_2) \ldots (k_n) \qquad \textbf{(4.11)}$$

- A state motor vehicle department would like to know how many license plate numbers are available if a license plate number consists of three letters followed by three numbers (0 through 9): (26)(26)(26)(10)(10)(10) = 17,576,000

- A restaurant menu has a price-fixed complete dinner that consists of an appetizer, an entrée, a beverage, and a dessert. You have a choice of 5 appetizers, 10 entrées, 3 beverages, and 6 desserts. Determine the total number of possible dinners.

- An online company is offering a date night special: pick one movie from four choices, one restaurant from six choices and either flowers, chocolates or wine. How many possible date night options are there?

Counting Rule 3

COUNTING RULE 3

The number of ways that all n items can be arranged in order is

$$n! = (n)(n-1)\ldots(1) \qquad \textbf{(4.12)}$$

where $n!$ is called n factorial, and $0!$ is defined as 1.

If a set of six books is to be placed on a shelf, in how many ways can the six books be arranged? n! = 6! = 6 x 5 x 4 x 3 x 2 x 1 = 720

Consider arranging 3 letters: A, B, C. How many ways can this be done?

Counting Rules 4

COUNTING RULE 4: PERMUTATIONS

The number of ways of arranging x objects selected from n objects in order is

$$ {}_nP_x = \frac{n!}{(n-x)!} \tag{4.13} $$

where

n = total number of objects
x = number of objects to be arranged
$n!$ = n factorial = $n(n-1)\ldots(1)$
P = symbol for permutations[1]

Counting Rule 4

If you have six books, but there is room for only four books on the shelf, in how many ways can you arrange these books on the shelf?

$$ {}_nP_x = \frac{n!}{(n-x)!} = \frac{6!}{(6-4)!} = \frac{(6)(5)(4)(3)(2)(1)}{(2)(1)} = 360 $$

How many ways can two slices of pizza be chosen from a plate containing one slice each of pepperoni, sausage, mushroom, and cheese pizza?

In a qualifying race for GGU road cycling event, 15 male MBA students competed for 5 slots. How many different ways could the top 5 slots be arranged if the order in which the men finished mattered?

Counting Rule 5

COUNTING RULE 5: COMBINATIONS

The number of ways of selecting x objects from n objects, irrespective of order, is equal to

$$ {}_nC_x = \frac{n!}{x!(n-x)!} \qquad (4.14) $$

where

n = total number of objects
x = number of objects to be arranged
$n!$ = n factorial = $n(n-1)\ldots(1)$
C = symbol for combinations[2]

Counting Rule 5

- How many possible 3 scoop combinations could you create at an ice cream parlor if you have 31 flavors to select from and no flavor can be used more than once in the 3 scoops?

- The total choices is n = 31, and we select X = 3.

$$ {}_{31}C_3 = \frac{31!}{3!(31-3)!} = \frac{31!}{3!28!} = \frac{31 \bullet 30 \bullet 29 \bullet 28!}{3 \bullet 2 \bullet 1 \bullet 28!} = 31 \bullet 5 \bullet 29 = 4{,}495 $$

Counting Rule 5

The number of combinations of selecting x objects out of n objects, in which order is not important, is:

$$ {}_{n}C_{x} = \frac{n!}{x!(n-x)!} $$

where

$n! = (n)(n-1)(n-2)\cdots(2)(1)$

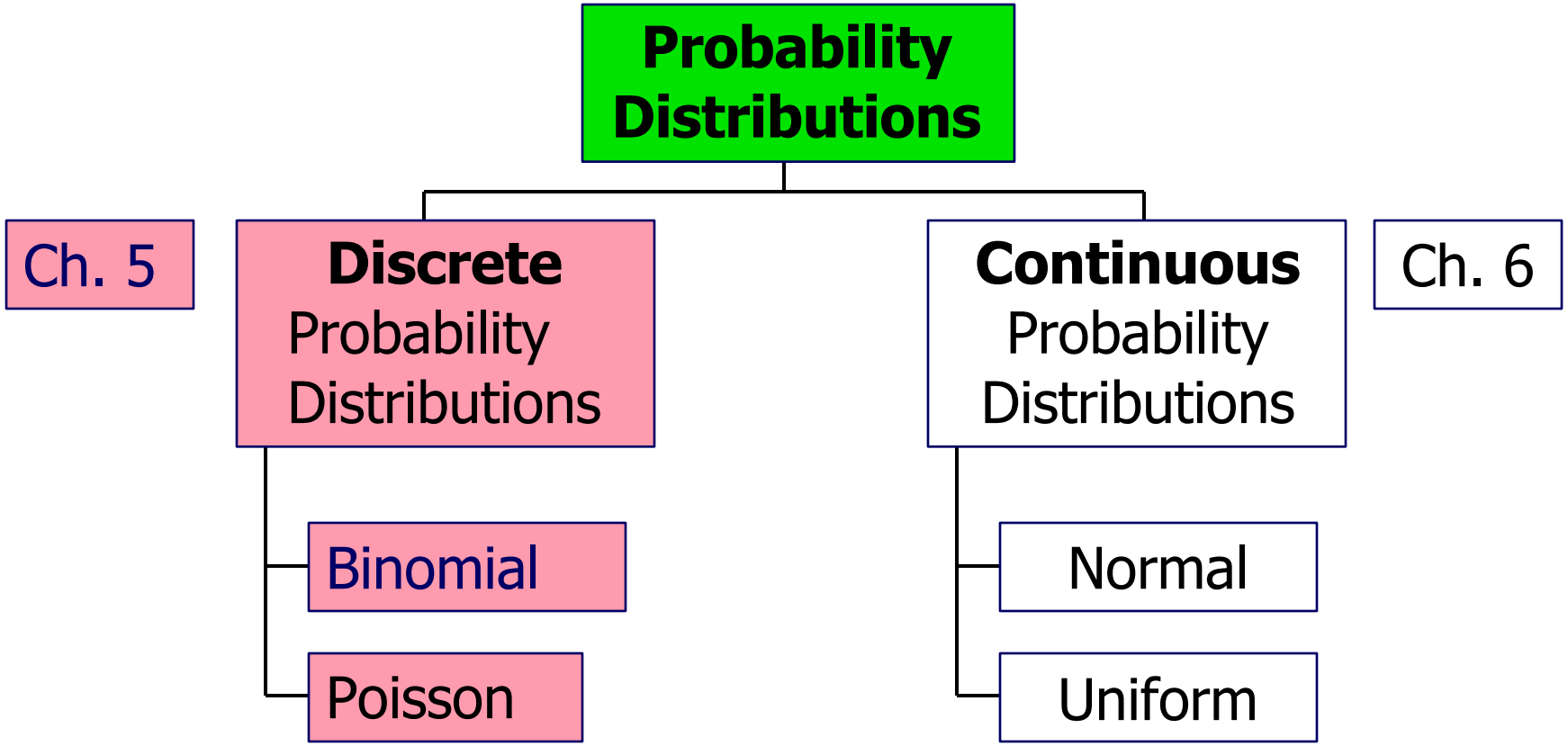
$x! = (X)(X-1)(X-2)\cdots(2)(1)$

$0! = 1$ (by definition)

Example Problems:

1. In how many ways can a coach choose three swimmers from among five swimmers?
2. How many different committees of 4 students can be chosen from a group of 15?
3. 10 people exchange greetings at a business meeting. How many greetings are exchanged if everyone greets each other once?
4. How many lottery tickets must be purchased to complete all possible combinations of six numbers, each with a possibility of being from 1 to 49?

Probability Distributions



Binomial Probability Distribution

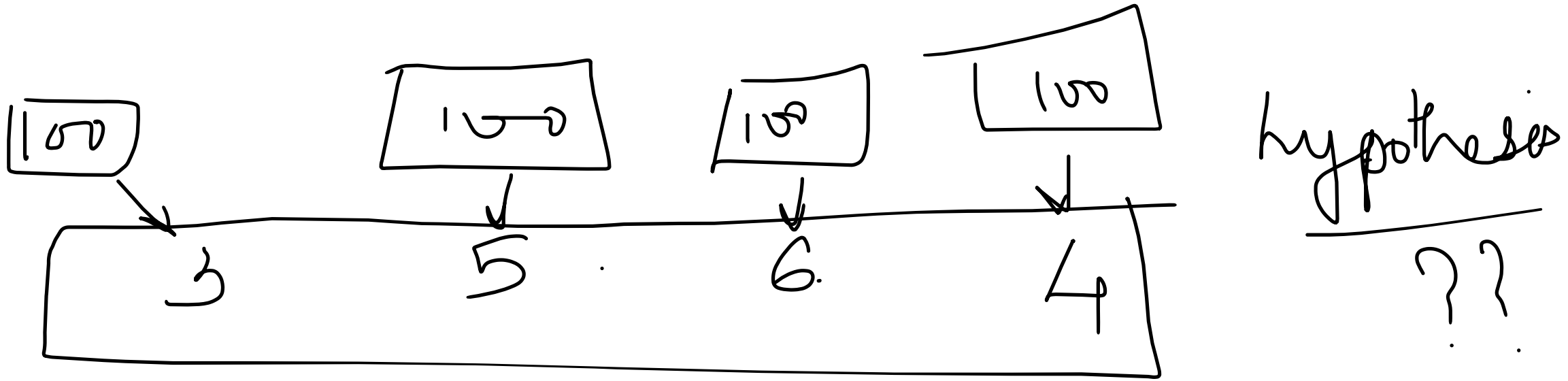
The distribution of the number of successes in a sequence of n independent experiments.

- A fixed number of observations, n
  - e.g., 15 tosses of a coin; ten light bulbs taken from a warehouse
- Each observation is categorized as to whether or not the "event of interest" occurred
  - e.g., head or tail in each toss of a coin; defective or not defective light bulb
  - Since these two categories are mutually exclusive and collectively exhaustive
    - When the probability of the event of interest is represented as π, then the probability of the event of interest not occurring is 1 - π

20000 $n = 100$ 0.04

Philips, Bayes Crompton

5 ✓ 6 4 1



hypotheses

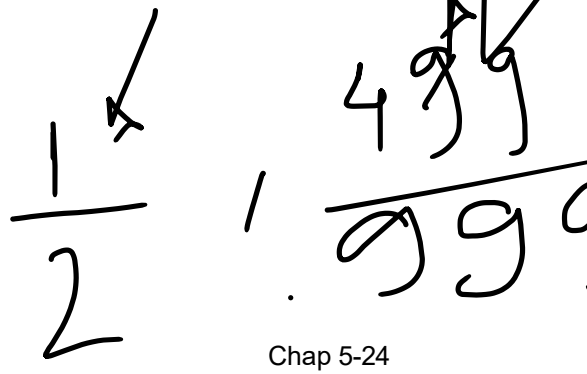
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Binomial Probability Distribution

- Constant probability for the event of interest occurring (π) for each observation
  - Probability of getting a tail is the same each time we toss the coin

- Observations are independent
  - The outcome of one observation does not affect the outcome of the other
  - Two sampling methods deliver independence
    - Infinite population without replacement
    - Finite population with replacement





Possible Applications for the Binomial Distribution

- A manufacturing plant labels items as either defective or acceptable
- A firm bidding for contracts will either get a contract or not
- A marketing research firm receives survey responses of "yes I will buy" or "no I will not"
- New job applicants either accept the offer or reject it

Binomial Distribution Formula

$$P(X=x \mid n,\pi) = \frac{n!}{x!\,(n-x)!}\,\pi^{x}\,(1-\pi)^{n-x}$$

$P(X=x|n,\pi)$ = probability that X = **x** events of interest, given **n** and **π**

x = number of "events of interest" in sample, (x = 0, 1, 2, ..., n)

n = sample size (number of trials or observations)

π = probability of "event of interest"

1 – π = probability of not having an event of interest

Example: Flip a coin four times, let x = # heads:

n = 4

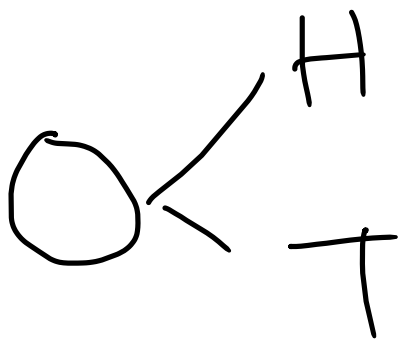
π = 0.5

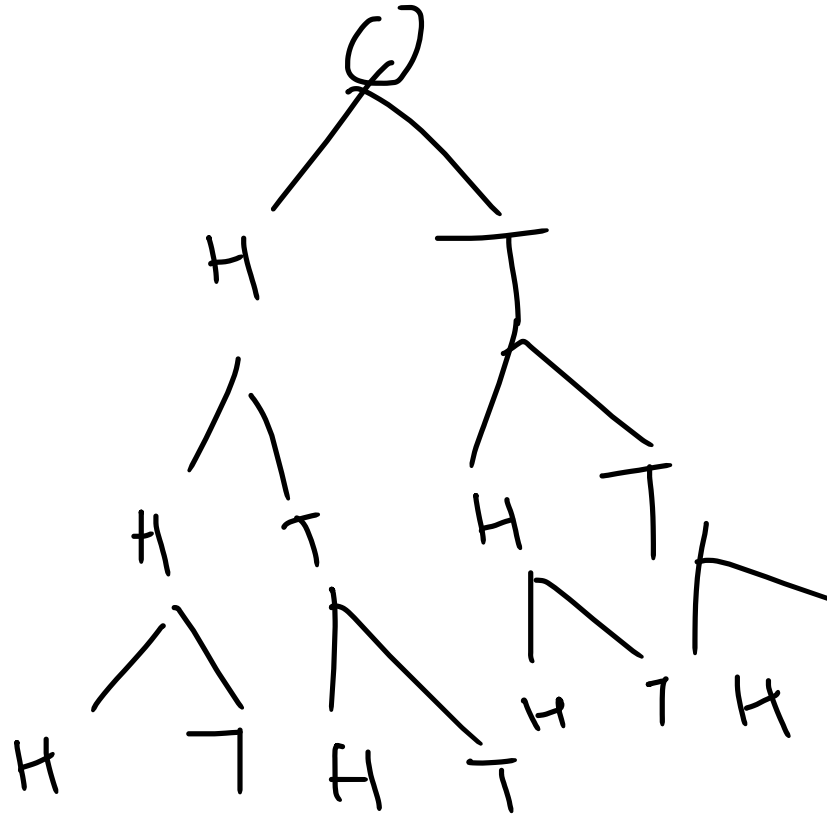
1 - π = (1 - 0.5) = 0.5

X = 0, 1, 2, 3, 4

$$^{n}C_{r}\, p^{r} q^{n-r}$$

$$^{n}C_{r}\, p^{r} (1-p)^{n}$$





Binomial Distribution Example

What is the probability of one success in five observations if the probability of an event of interest is 0.1?

x = 1, n = 5, and π = 0.1

$$P(X=1|\,5,0.1) = \frac{n!}{x!(n-x)!}\pi^{x}(1-\pi)^{n-x}$$

$$= \frac{5!}{1!(5-1)!}(0.1)^{1}(1-0.1)^{5-1}$$

$$= (5)(0.1)(0.9)^{4}$$

$$= 0.32805$$

Binomial Distribution Example

Suppose the probability of an invoice payment being late is 0.10. What is the probability of 1 late invoice payment in a group of 4 invoices?

x = 1, n = 4, and π = 0.10

$$P(X=1|\,4, 0.10) = \frac{n!}{x!(n-x)!}\pi^{x}(1-\pi)^{n-x}$$

$$= \frac{4!}{1!(4-1)!}(0.10)^{1}(1-0.10)^{4-1}$$

$$= (4)(0.10)(0.729)$$

$$= 0.2916$$

Using Excel

	A	B	
1	**Binomial Probabilities**		
2			
3	**Data**		
4	Sample size	4	
5	Probability of an event of interest	0.1	
6			
7	**Statistics**		
8	Mean	0.4	=B4 * B5
9	Variance	0.36	=B8 * (1 - B5)
10	Standard deviation	0.6	=SQRT(B9)
11			
12	**Binomial Probabilities Table**		
13	***X***	***P(X)***	
14	0	0.6561	=BINOM.DIST(A14, B4, B5, FALSE)
15	1	0.2916	=BINOM.DIST(A15, B4, B5, FALSE)
16	2	0.0486	=BINOM.DIST(A16, B4, B5, FALSE)
17	3	0.0036	=BINOM.DIST(A17, B4, B5, FALSE)
18	4	0.0001	=BINOM.DIST(A18, B4, B5, FALSE)

Suppose the probability of an invoice payment being late is 0.10. What is the probability of 1 late invoice payment in a group of 4 invoices?

$x = 1$, $n = 4$, and $\pi = 0.10$

The Poisson Distribution

The Poisson Distribution Definitions

- You use the **Poisson distribution** when you are interested in the number of times an event occurs in a given area of opportunity.

- An **area of opportunity** is a continuous unit or interval of time, volume, or such area in which more than one occurrence of an event can occur.

- It is a discrete probability distribution of the number of events occurring in a given time period, given the average number of times the event occurs over that time period.

The Poisson Distribution Definitions

- You use the **Poisson distribution** when you are interested in the number of times an event occurs in a given area of opportunity.

- An **area of opportunity** is a continuous unit or interval of time, volume, or such area in which more than one occurrence of an event can occur.

Example of Areas of Opportunity

- The number of scratches in a car's paint
- The number of mosquito bites on a person
- The number of computer crashes in a day
- The number of traffic accidents at a particular intersection
- The number of house fire claims per month that are received by an insurance company
- The number of people who are infected with the AIDS virus in a certain neighborhood
- The number of people who walk into a barber shop without an appointment

The Poisson Distribution

- Apply the Poisson Distribution when:
  - You wish to count the number of times an event occurs in a given area of opportunity (specific interval of time or in a specific distance, area, or volume)
  - The probability that an event occurs in one area of opportunity is the same.
  - Each event is independent of all other events.
- The average number of events per unit is λ (lambda)

Poisson Distribution Formula

$$P(X = x \mid \lambda) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

where:

x = number of events in an area of opportunity

λ = expected number of events

e = base of the natural logarithm system (2.71828...)

The Poisson Distribution Example

The average customer arrival rate in a 10 minute window at a fast food drive up window is 0.5 customers per minute. What is the probability that 2 customers arrive in a minute? x = 2, λ = 0.50

$$P(X=2\mid 0.5)=\frac{e^{-\lambda}\lambda^{x}}{x!}=\frac{e^{-0.50}\,0.50^{2}}{2!}$$

$$=\frac{0.606531*0.25}{2}$$

$$=0.0758$$

Using Poisson Tables (Available Online)

X	λ								
	0.10	0.20	0.30	0.40	**0.50**	0.60	0.70	0.80	0.90
0	0.9048	0.8187	0.7408	0.6703	0.6065	0.5488	0.4966	0.4493	0.4066
1	0.0905	0.1637	0.2222	0.2681	0.3033	0.3293	0.3476	0.3595	0.3659
2	0.0045	0.0164	0.0333	0.0536	**0.0758**	0.0988	0.1217	0.1438	0.1647
3	0.0002	0.0011	0.0033	0.0072	0.0126	0.0198	0.0284	0.0383	0.0494
4	0.0000	0.0001	0.0003	0.0007	0.0016	0.0030	0.0050	0.0077	0.0111
5	0.0000	0.0000	0.0000	0.0001	0.0002	0.0004	0.0007	0.0012	0.0020
6	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0001	0.0002	0.0003
7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Example: Find $P(X = 2 \mid \lambda = 0.50)$

$$P(X=2|0.50)=\frac{e^{-\lambda}\lambda^{X}}{X!}=\frac{e^{-0.50}(0.50)^{2}}{2!}=0.0758$$

Using Excel For The Poisson Distribution

	A	B	C	D	E
1	**Poisson Probabilities**				
2					
3	**Data**				
4	Mean/Expected number of events of interest:				0.5
5					
6	**Poisson Probabilities Table**				
7	**X**	**P(X)**			
8	0	0.6065	=POISSON.DIST(A8,E4,FLASE)'		
9	1	0.3033	=POISSON.DIST(A9,E4,FLASE)'		
10	2	0.0758	=POISSON.DIST(A10,E4,FLASE)'		
11	3	0.0126	=POISSON.DIST(A11,E4,FLASE)'		
12	4	0.0016	=POISSON.DIST(A12,E4,FLASE)'		
13	5	0.0002	=POISSON.DIST(A13,E4,FLASE)'		
14	6	0.0000	=POISSON.DIST(A14,E4,FLASE)'		
15	7	0.0000	=POISSON.DIST(A15,E4,FLASE)'		
16	8	0.0000	=POISSON.DIST(A16,E4,FLASE)'		
17					

Chapter Summary

In this chapter we discussed

- The probability distribution of a discrete random variable
- The Binomial distribution
- The Poisson distribution

Reminders for Session 6

- READ

  Chapter 6, Section 6.1 – 6.2.

- REVIEW the EXAMPLES

  Download the examples. If you have trouble following any of the examples, ask your instructor or a math tutor for help.

- DO the QUIZ

  You got two attempts and the higher score is recorded.

- DO the HOMEWORK

  This is due on or before 11:55PM of February 11.